



COURSE CONTENT AND PLANNING	
SEMESTER-VI	
Course Name: Numerical Methods and its Applications	Course Code: BMANM655
No. of Lecture hours / week: 3	CIE Marks: 50
No. of Tutorial/Practical hours / week: 0	SEE Marks: 50
Total No. of Lecture Hours + Tutorial/Practical hours: 40	SEE Duration: 3 Hrs
L: T: P: 3:0:0	Credits: 3
COURSE CONTENT	
Module I: Roots of Polynomial Roots of Polynomials: Bairstow's method, Graeffe's Roots Squaring Method. SLT: Approximations and round off errors: Significant figures, accuracy and precision, error definitions, roundoff errors and truncation errors.	
Module II: Solution of Linear System - Direct and Iterative Methods Matrix Partition method, Positive Definite matrix and Cholesky decomposition SLT: Least Square method for Inconsistent systems.	
Module III: Numerical Differentiation and Numerical Integration Numerical Differentiation (Newton Forward, backward and divided difference). Romberg integration and double integral using Simpson's (1/3)rd rule. SLT: Maxima and Minima using interpolation	
Module IV: Eigen Values and Eigen Vectors of Symmetric matrices Eigen Values and Eigen Vectors of symmetric matrices using Jacobi and Givens method for symmetric matrices, Householder's method for symmetric matrices. SLT: Rutishauser method for arbitrary matrices.	
Module V: Numerical solution for IVP and BVP IVP: Picard's method (first order) BVP: Successive Over Relaxation method and Alternating Direction Implicit method. SLT: Picard's method (second order)	
Text Books 1. S.S.Sastry, Introductory Methods of Numerical Analysis, PHI, 2005. 2. Steven C. Chapra, Raymond P. Canale, Numerical Methods for Engineers, Tata McGraw Hill, 4th Ed., 2002. 3. M K Jain, S.R.K. Iyengar, R K Jain, Numerical Methods, for Scientific and Engg. Computation, New Age International, 2003. 4. Madhumangal Pal, Numerical Analysis for Scientists and Engineers, Narosa Publishing House, Latest Edition.	
Reference Books: 1. Pervez Moin, Fundamentals of Engineering Numerical Analysis, Cambridge, 2010.	



Vidyavardhaka Sangha[®], Mysore VIDYAVARDHAKA COLLEGE OF ENGINEERING

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(Approved by AICTE, New Delhi & Government of Karnataka)

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2. David C. Lay, Linear Algebra and its Applications, 3rd edition, Pearson Education, 2002.

Course Outcomes: At the end of this course, students will be able to

C01	Understand the basic concepts of Roots of a polynomial, Solution of a Linear system, Numerical differentiation and integration, Eigen values and Eigen vectors of Symmetric matrices, Boundary Value Problems in Ordinary and Partial differential equations. (PO-1) .
C02	Apply the concept of Linear system, Numerical differentiation and integration, Eigen values and Eigen vectors of Symmetric matrices, Boundary Value Problems in Ordinary and Partial differential equations (PO-1) .
C03	Analyse the concept of Linear system, Numerical differentiation and integration, Eigen values and Eigen vectors of Symmetric matrices, Boundary Value Problems in Ordinary and Partial differential equations (PO-2) .
C04*	Use modern tool to solve/visualize mathematical problems arising in Engineering (PO-5) .

SEE - Course Assessment Plan

CO	Marks Distribution					Total Marks	Weightage (%)
	Module-1	Module-2	Module-3	Module-4	Module-5		
C01	10	-	-	--	--	10	10%
C02	10	10	8	20	20	68	68%
C03	-	10	12	-	-	22	22%
	20	20	20	20	20	100	100%

Intra-Module Choice for Answering of Questions

Module - III

Module - IV

In other modules there will be no choice for answering